Lecture 4: Electric Field Lines

Pre-Lecture Worksheet

Complete the following worksheet in preparation for the next lecture. The following questions correspond to material covered in chapter 21. Feel free to read chapter 21 and answer the questions. You may also seek answers to the questions from other sources. Note: the textbook will be used as the final source for these questions. While answers gleaned from other sources may be correct, answers must encompass the material covered in the book to be considered correct.

Electric field lines are a way of visualizing the magnitude and direction of an electric field without having to calculate these values at each and every point. Answer the following questions to better understand the rules of drawing field lines. When appropriate, circle the word that makes the statement true.

1) (True or False) The tangent of a field line at any point is the direction of the electric field at that location.

   true

2) Electric fields start at (positive or negative) charges and point to (positive or negative) charges.

3) Field lines are radial, close to a (surface or point).

4) (True or false) The density of field lines is proportional to the magnitude of the field line.

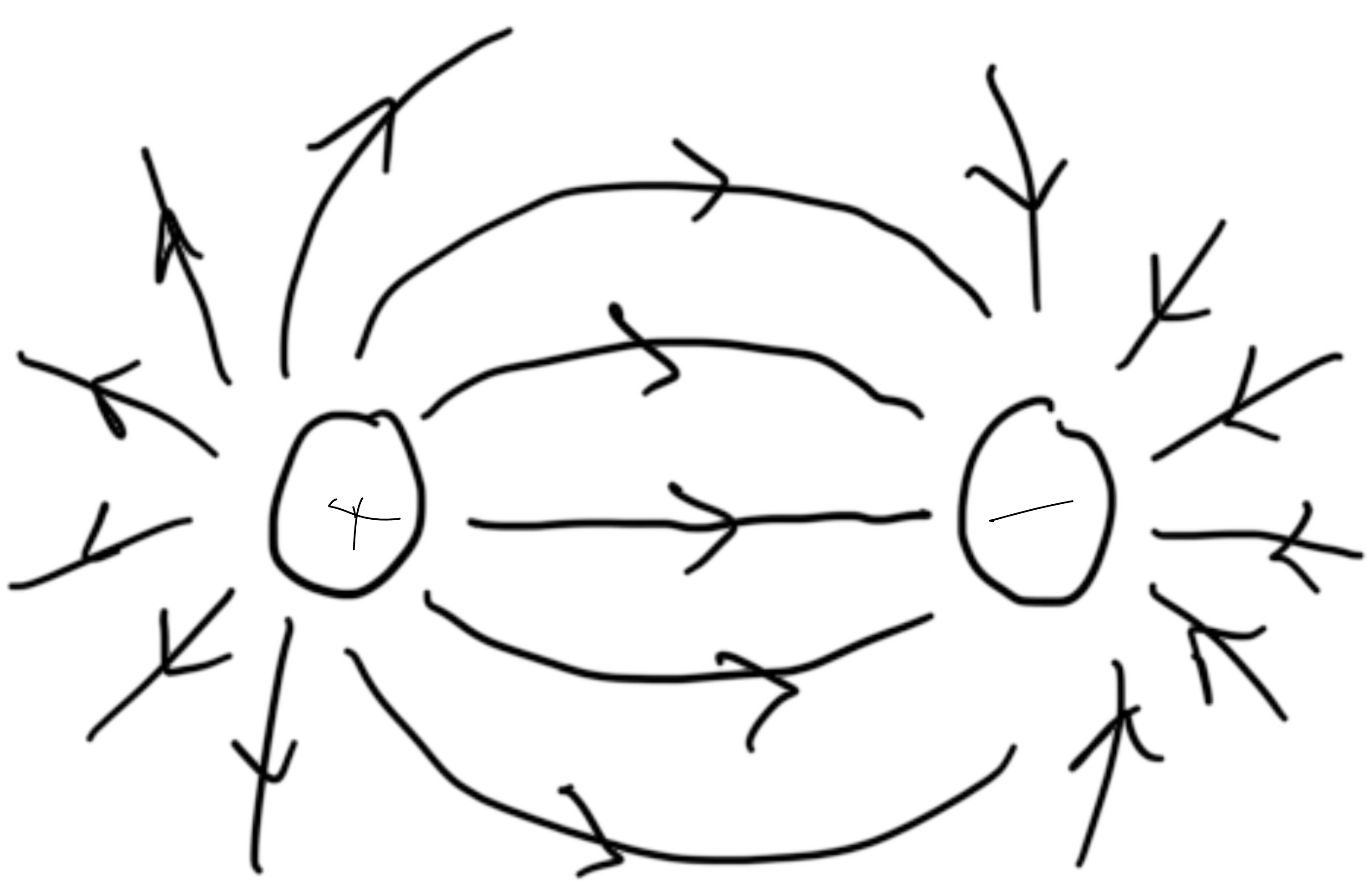
5) (True or false) The number of field lines corresponds directly to the charge of a point charge.

6) (True or false) Field lines can cross each other.

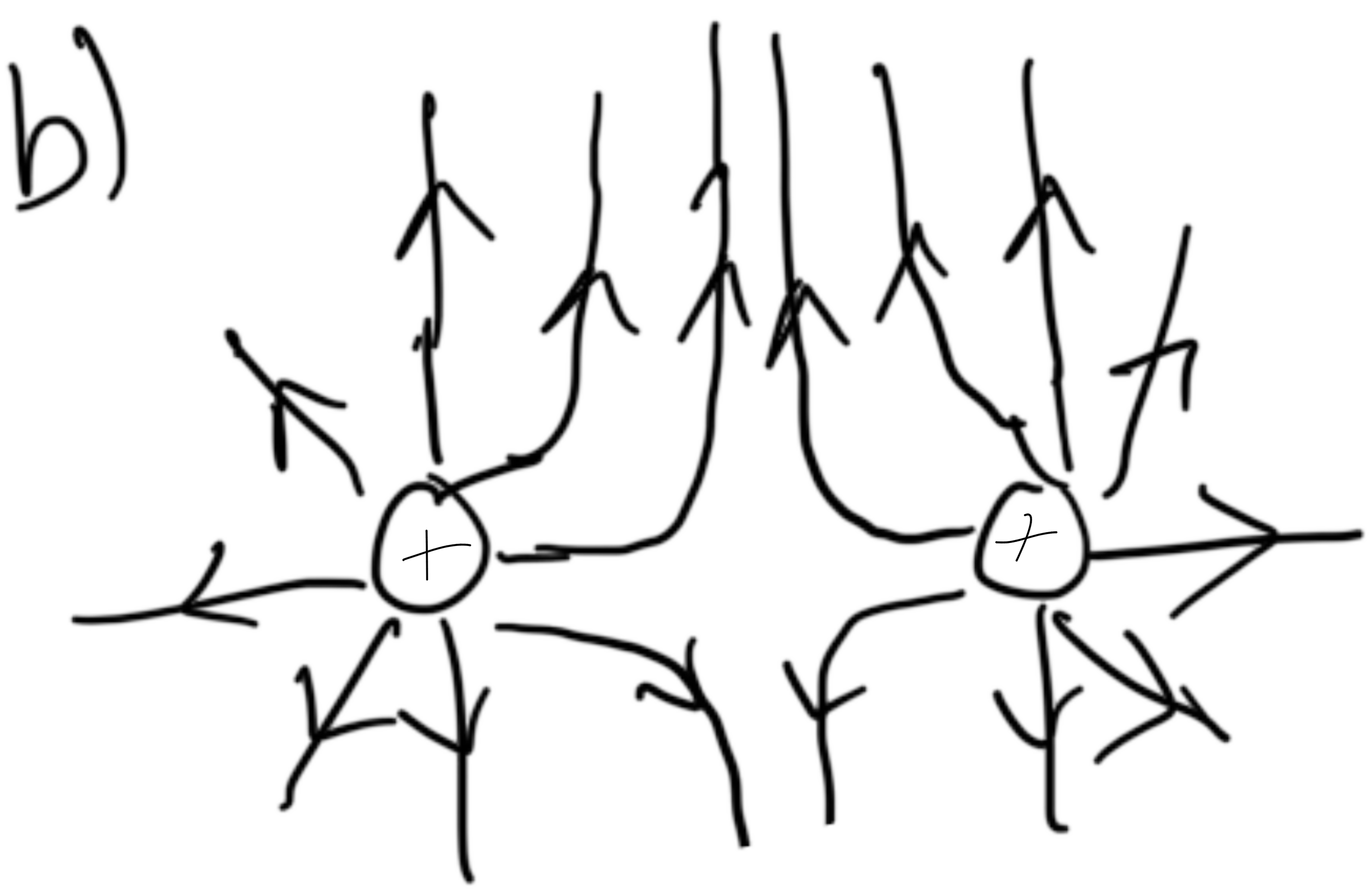
7) Arrows indicate the direction of the field and show up in the middle of the line. An arrow should point (toward or away from) a positive charge and (toward or away from) a negative charge.

8) Look at the following field line cases and label which charges are positive and which charges are negative.

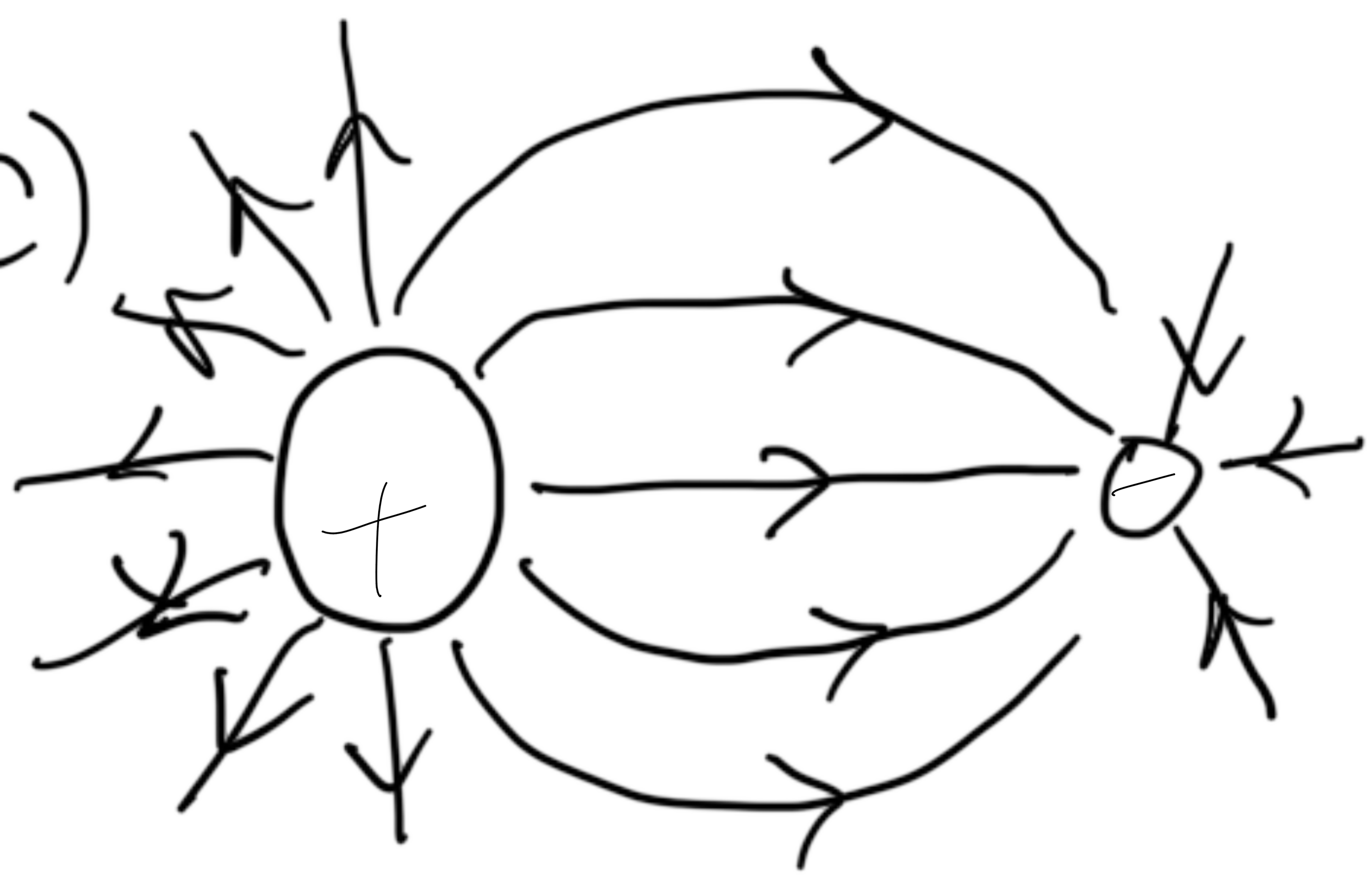
a)



b)



c)



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